

Draftly

*A Lawyer-in-the-Loop Legal Document Drafting and
Review Platform for Sri Lankan Legal Workflows*

Group 06

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1. Introduction

In Sri Lanka, many legal workflows still depend on repeated manual drafting and review. Lawyers and legal clerks prepare documents such as deeds, title reports, affidavits, petitions, legal notices, certificates and letters by reading supporting material, extracting the relevant facts, checking them, and then adapting previous drafts or templates. The difficult part is not only typing the final document; it is identifying the correct facts from client-specific records and ensuring that the draft is complete, consistent and ready for professional review.

Over the last few years, several Sri Lankan legal platforms have appeared, such as LawLanka, AI Pazz, Paralegal.lk and LawCeylon [1, 2, 3, 4]. These primarily support legal research, case law access and legislation retrieval, and they are widely used in practice. Draftly is designed to complement these tools rather than replace them: it focuses on client-specific legal document drafting and review, where the required content must be derived from the user's own uploaded documents and checked by a lawyer.

For the first use case, we focus on conveyancing and title-documentation workflows. This is a suitable starting point because lawyers must derive outputs such as title reports, pedigrees, abstracts of title, ownership certificates and assessor letters from prior deeds, survey plans and related records. It also gives the project a clear data science component, because facts must be extracted and linked across documents before any draft can be generated.

To confirm that this gap is real, we have conducted preliminary discussions with practicing Sri Lankan lawyers, notaries and legal academics specialising in conveyancing. Their feedback indicates that while legal research platforms are useful, there remains a practical gap in preparing repeated legal documents from client-specific records. This is the gap Draftly is built to explore.

In short: Draftly is a lawyer-in-the-loop drafting and review platform that turns uploaded legal documents and verified facts into reviewable drafts. In the first use case, it applies this idea to conveyancing and title documentation.

2. Proposed Solution

Draftly is a *lawyer-in-the-loop* platform: the system proposes and the lawyer reviews, edits and approves. No output is ever treated as a final legal document on its own; every draft is presented for a qualified professional to check and sign off on.

In a typical workflow, the lawyer or legal clerk creates a matter, uploads supporting documents,

reviews the facts suggested by the system, corrects them where necessary and then generates an editable draft from an approved template. For the initial conveyancing use case, the supporting documents may include previous deeds, survey plans and assessment records, and the outputs may include a title report, pedigree, abstract of title, ownership certificate or assessor letter.

2.1 Research Novelty

The template is only the final output layer. The more important part is the earlier document-understanding workflow: reading uploaded material, extracting structured facts, linking related facts across documents, identifying missing or inconsistent information, and showing the lawyer where each fact came from.

This makes Draftly different from a simple form-filling system. In a form-filling system, the user manually types all values into text boxes and the system places them in a document. In Draftly, the system attempts to derive candidate facts from uploaded documents, presents them with evidence, and lets the lawyer verify or correct them before a draft is produced. In the conveyancing use case, this includes tasks such as extracting deed numbers, parties, dates, plan numbers, lot numbers and extents, then using those facts to support title reports, pedigrees and related documents.

2.2 Key Features

- Upload and organise legal documents for a matter
- Extract important facts from scanned and digital legal documents
- Show extracted facts with source evidence for lawyer verification
- Highlight missing or inconsistent information
- Generate editable drafts from approved templates and verified facts
- Support lawyer review, correction, approval and export

2.3 Data Science Component

The data science component focuses on document understanding and fact extraction from Sri Lankan legal documents. For the first use case, this includes OCR for scanned deeds and forms, extraction of parties, deed numbers, dates, notaries, property details, registration references and survey-plan details, and linking those facts across a small document bundle. The project can compare suitable extraction approaches and evaluate them against annotated examples.

The conveyancing use case also allows limited title reasoning. Once deed and property facts are extracted, the system can help arrange the ownership history, identify missing documents and flag possible inconsistencies such as name variations, plan-number mismatches or extent differences. These checks are presented as support for lawyer review, not as final legal conclusions.

2.4 Software Engineering Component

The software engineering component is the lawyer-facing workflow around the models: matter creation, document upload, fact review, draft generation, editing and export. The exact implementation choices can be decided later, but the project idea requires a usable prototype that demonstrates the end-to-end flow from uploaded documents to lawyer-reviewable drafts. Because legal documents are sensitive, privacy, access control and auditability will be considered from the design stage.

2.5 Evaluation and Objectives

The system will be evaluated using measurable criteria:

- Accuracy of document classification and key-field extraction
- Correctness of linked facts and simple title-chain support in the first use case
- Quality of missing-field and inconsistency warnings
- Lawyer assessment of generated drafts and source traceability
- Time saved compared with fully manual preparation

3. Datasets

There is no ready-made public dataset for Sri Lankan legal document drafting workflows, so building a small project dataset is part of the work. For the first use case, data sources include anonymised historical deeds, title reports, survey plans, assessor letters, ownership certificates and lawyer-approved drafting templates, collected with permission from supporting practitioners. For general legal reference material, we can refer to publicly available or subscription-based sources such as LawLanka [1], where appropriate.

On top of the raw documents, we will build a small annotated dataset that labels entities such as parties, dates, deed numbers, boundaries, extents and plan references, together with selected links between related documents. All documents will be anonymised before use and handled under clear consent from the professionals who share them.

4. Similar Projects

Local. LawLanka, AI Pazz, Paralegal.lk and LawCeylon [1, 2, 3, 4] are among the closest Sri Lankan systems. Based on our preliminary review, these platforms mainly focus on legal search, case law, legislation access and related research support.

International. Tools such as Harvey AI, Spellbook, Lexis+ AI and CoCounsel [5, 6, 7, 8] show that AI-assisted legal drafting is both viable and commercially valuable. However, these are mostly built around other jurisdictions and legal workflows, so they do not directly address the needs of Sri Lankan document drafting and review.

Related research Since Draftly is in the Sri Lankan legal NLP domain, we will also review the previous work carried out under the SigmaLaw project by Dr. Nisansa de Silva’s research group [9]. SigmaLaw has explored legal information extraction, legal party identification, named entity recognition, coreference resolution, semantic similarity, sentence embeddings, and legal information retrieval, which are directly relevant to Draftly’s broader goal of understanding legal documents before drafting or review. The most relevant papers for our project include “Party Identification of Legal Documents using Co-reference Resolution and Named Entity Recognition” [10], “Legal Party Extraction from Legal Opinion Texts Using Recurrent Deep Neural Networks” [11], “Legal Party Extraction from Legal Opinion Text with Sequence to Sequence Learning” [12], “Context Sensitive Verb Similarity Dataset for Legal Information Extraction” [13], “Learning Sentence Embeddings in the Legal Domain with Low Resource Settings” [14], and “Legal Document Retrieval using Document Vector Embeddings and Deep Learning” [15]. We will use these works to understand existing approaches for legal entity extraction, party linking, legal semantic representation and information retrieval. Draftly differs by applying such legal NLP ideas inside a lawyer-in-the-loop drafting and review product: the system organises a client’s uploaded legal documents, extracts and links facts with source evidence, highlights missing or inconsistent information, and generates editable drafts for professional review. Conveyancing and title documentation will be used as the first practical workflow to validate this broader idea.

How Draftly differs. Draftly is positioned as a complementary system for client-specific

legal document drafting and review. Instead of only retrieving legal sources, it helps organise uploaded matter documents, extract relevant facts, present them for lawyer verification and generate editable drafts. Conveyancing and title documentation are used as the first concrete workflow to prove this idea.

5. Scope and Team Responsibilities

The project is scoped so that both a data-science and a software-engineering contribution are clearly present and so that each team member owns a distinct part of the system:

- **Document understanding (Data Science):** OCR support, entity extraction and comparison of extraction approaches.
- **Fact linking and checking (Data Science):** linking related facts across documents and identifying missing or inconsistent information.
- **Draft generation (Data Science / SE):** producing reviewable drafts from verified facts and approved templates.
- **Review workflow (Software Engineering):** matter management, document upload, fact review, draft editing and export.

For the project idea, Draftly is broader than conveyancing, but the MVP is deliberately limited to conveyancing and title-documentation workflows such as title reports, pedigree generation, abstracts of title, ownership certificates and assessor letters. After this first workflow is proven, the same product direction can be extended to complaints, affidavits, petitions, legal notices and other Sri Lankan legal documents.

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